Modern Graphic Technologies on Mali GPUs

> 1.2 Bn GPU shipped to date

- Multiple product
  - Mali G3x → entry-level
  - > Mali G5xx → mid-end (balance performant left)
  - > Mali G7xx → sub-flagship
  - > Mali Immortalis → flagship

Immortalis_G925

- CSF replacing JM
  - > GPU can now micro managed command with dedicated MCU
- DVS
  - > less memory bandwidth an power
- HSR
  - > surface removal

History of Mali GPU

2021 → 2025

Trend in mobile gaming

Immortalis adopts

- PBR shading
- vivid motion blur
- high polygon asset
- ray tracing

Intr. Ray Tracing on Mobile

- ada trace → select hits → color pixels
- determining what appears on screen
- ray tracing itu ~~merefleksikan~~ untuk meningkatkan realistis dari lighting

Caranya

- miss object ditampilkan
- hit object shadows
- angle menentukan light intensity

Ray-tracing techniques

- basic lighting
- basic shadowing
- indirect reflections
  - > susah untuk rasterizer
- refractions
  - > susah juga untuk rasterizer
- soft shadows
  - > susah juga untuk rasterizer
- indirect illumination

need lots of rays

- solusinya
  - > denoiser
  - > monte carlo sampling
  - > importance sampling

Use machine learning to ~~face~~ realtime ray tracing' need less ~~the~~ rays

- denoise
- anti-aliasing
- upscaling
- frame generation

Ray tracing in API

Ray query, mampu

- untuk membangun acceleration structure
- cast rays perform
- intersection test

shader query

↓

Ray tracing pipeline

- gunakan hardware untuk traversal, ray scheduling, shader dispatch
- lalu shade

Core Traverse ~~pitaem~~ triangle

- use hierarchical data structure

Acceleration structure

- Jadi ada tree contains geometry, kalau miss, skip branch true

- ada dua level
  - > top level contains scene composition
  - > bottom level generate object based on top level

Ada best practice untuk construction dan traversal
~~di produk~~

Di produk ARM, mereka bagi bagus
- mana rasterized
- mana ray tracing

Ray traction unit ada komponen-komponen di dalamnya:

Arm Neural Technology
- neural graphics : ML + graphics rendering pipeline
  - to higher quality & rendering cost
    real time rendering ✓
- value
  - > permanen
  - > eff
  - > guna
  - > future proof
- bukan pp npu, tapi sekitas, dan embedded di GPU

- use case
  - > neural super sampling
  - > neural frame rate upscaling
  - > neural super sampling and denoising

Super resolution

- render low resolution and then use upscaling / use neural / spatial / neural to upscale to get higher resolution

Spatial Upscaling

- hanya melihat satu frame, jadi lemah kalau ada pergerakan

Temporal Upscaling

- lebih bagus dari spatial, tapi jauh lebih complex

Neural Upscaling

- predict and generate.
- cons : training, optimization

NSS, kebanyak open model

- inputs : color, depth, motion

NSS seperti NFRU dan NRSD